PRIYANKA BOLEM

MACHINE LEARNING ENGINEER

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SUMMARY

ML Engineer with 4+ years building and shipping production models over high-volume, real-time data. Depth in predictive and probabilistic modeling, uncertainty estimation, and large-scale feature engineering, paired with direct ownership of low-latency inference on AWS SageMaker, distributed pipelines in Spark, Databricks, and Airflow, and production monitoring for drift and model decay. MS in Computer Science; strong applied grounding in Python, SQL, PyTorch, TensorFlow, and experimentation at scale.

TECHNICAL SKILLS

Languages: Python, SQL, R

ML and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Random Forest, Hugging Face Transformers, OpenCV, NLP, Vector Databases

Modeling and Optimization: Regression, Classification, Clustering, Neural Networks, Bayesian Optimization, Monte Carlo Simulation, Genetic Algorithms, Reinforcement Learning, Hyperparameter Tuning, Model Calibration, Uncertainty Estimation

Data and Pipelines: Apache Spark, Kafka, Airflow, Azure Databricks, ETL design, Feature Engineering, DVC

MLOps and Deployment: AWS SageMaker (Pipelines, Endpoints), MLflow, Docker, Kubernetes, FastAPI, GitHub Actions, CI/CD, CloudWatch, Evidently, Prometheus, Grafana, Drift Detection

Cloud: AWS (SageMaker, Lambda, EC2, ECS, S3), Azure (ML Studio, Databricks)

Evaluation and Experimentation: A/B Testing, Cross-validation, ROC-AUC, Precision/Recall/F1, AUPRC, Model Calibration

Generative AI and NLP: GPT-4, BERT, FinBERT, LLaMA, LangChain, LangGraph, RAG, Prompt Engineering, Tokenization, Neo4j

Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Pandas, NumPy, Statsmodels

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Truveta *Seattle, WA | Jun 2025 - Present*

- Deployed end-to-end ML pipelines on AWS SageMaker, using SageMaker Pipelines for training automation, SageMaker Endpoints for real-time risk scoring, and CloudWatch for production monitoring and alerting, ensuring reproducibility and compliance with healthcare data standards.
- Designed and implemented predictive models in Python and TensorFlow to forecast chronic disease progression from de-identified EHR and demographic data, enabling automated patient stratification for clinical research and population health studies.
- Applied Bayesian Optimization and Monte Carlo simulation across large patient cohorts to quantify uncertainty in outcome predictions, improving prediction precision and providing data-driven insight into intervention effectiveness.
- Built a hybrid optimization framework pairing Genetic Algorithms for cohort selection with Reinforcement Learning for adapting longitudinal strategies, integrating historical records with real-time signals for continuous learning; improved plan efficiency and outcome prediction accuracy by 22%.
- Encoded high-cardinality unstructured text (lab reports, discharge summaries, medication history) as BERT embeddings, enriching model inputs with contextual signal from raw clinical language.
- Optimized SQL queries and data pipelines over large-scale datasets in a cloud-native data lake, reducing model training and data preparation time by 35% and accelerating delivery for downstream data science and research teams.

- Partnered with research and data product teams to build Power BI dashboards tracking incidence trends, treatment efficacy across subpopulations, and regional risk distribution, supporting resource allocation across health systems.

Machine Learning Engineer | Capgemini *India | Apr 2019 - Nov 2022*

- Built and deployed models with Scikit-learn, XGBoost, and LightGBM to predict credit default probability, loan delinquency, and fraudulent claims, then embedded them into origination, collections, and fraud triage systems with risk and compliance partners, reducing end-to-end loan decision cycle time by 25%.
- Engineered features from large-scale financial, credit bureau, and behavioral data using Pandas and NumPy, capturing payment patterns, credit utilization, delinquency history, and spending trends.
- Built automated ETL pipelines in Python and SQL across core banking systems, credit bureaus, and CRM platforms, reducing data preparation time by 40%; scaled preprocessing, feature engineering, and distributed training on Azure Databricks.
- Orchestrated Apache Airflow workflows for scheduled retraining, data refresh, and batch inference, keeping predictions accurate as customer behavior and regulatory requirements changed.
- Developed a model performance monitoring system in Python tracking accuracy, drift, and alert thresholds, reducing false positives by 20% and improving early warning signals across credit and fraud models.
- Designed interactive Tableau dashboards for credit risk analysts, fraud teams, and loan officers to monitor approval rates, default probabilities, and fraud detection rates across customer segments, improving reporting efficiency by 18%.

PROJECTS

- **Financial Sentiment MLOps Platform.** End-to-end NLP platform: FinBERT fine-tuning, multi-source ingestion, MLflow tracking, FastAPI + Docker + Kubernetes serving, Prometheus/Grafana monitoring, Evidently drift detection, GitHub Actions CI/CD; live demo on Hugging Face Spaces. github.com/priyankabolem/financial-sentiment-mlops • live demo
- **Credit Card Fraud Detection: Production MLOps.** Real-time fraud detection on a 1:578-imbalanced dataset (AUPRC-driven evaluation), XGBoost, MLflow registry, Airflow orchestration, Evidently drift monitoring, FastAPI on AWS ECS Fargate, and an LLM (Claude API) RAG layer for fraud-pattern explainability. github.com/priyankabolem/fraud-detection-mlops
- **RAG Chatbot (Neo4j knowledge graph + vector retrieval).** Retrieval-augmented assistant combining a Neo4j knowledge graph with OpenAI-embedding vector search (hybrid retrieval) through LangChain, deployed on Streamlit Cloud. github.com/priyankabolem/RAG-Chatbot-1 • live demo
- **Plant Disease Detection.** TensorFlow/Keras CNN image classifier across 38 disease classes with Grad-CAM explainability, Dockerised Streamlit app and CI; deployed to Hugging Face Spaces. github.com/priyankabolem/PlantDiseasesDetection • live demo
- **Lending Club Loan Default Prediction.** Credit-risk classifier on ~38k loans (LR, RF, XGBoost) with class-imbalance handling and threshold tuning; ROC-AUC > 0.85. github.com/priyankabolem/lendingclub-loan-approval-ml

EDUCATION

Master of Science, Computer Science | Northwest Missouri State University, Maryville, MO *Jan 2023 - May 2025*